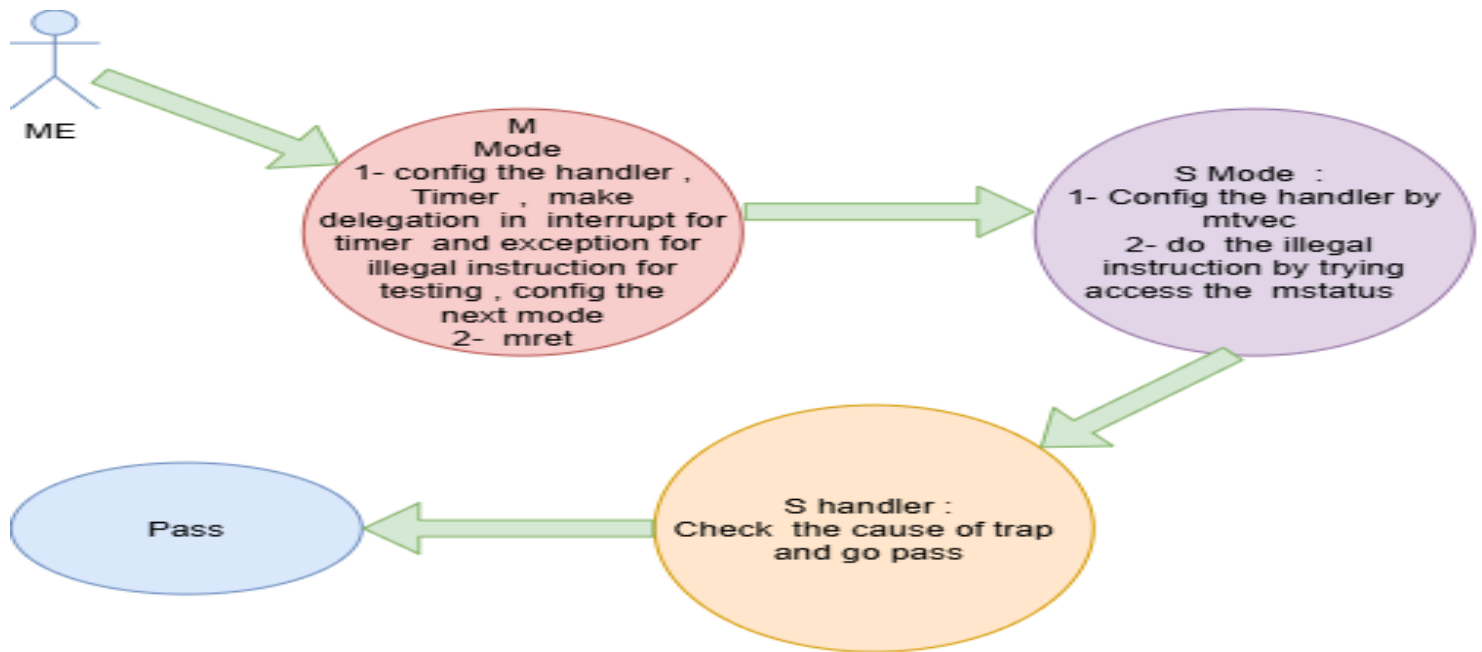


LAB ASSIGNMENT: Hardware Timer Delegation Across M-Mode, S-Mode, and U-Mode

Task1 :

Diagram Flow Testing :

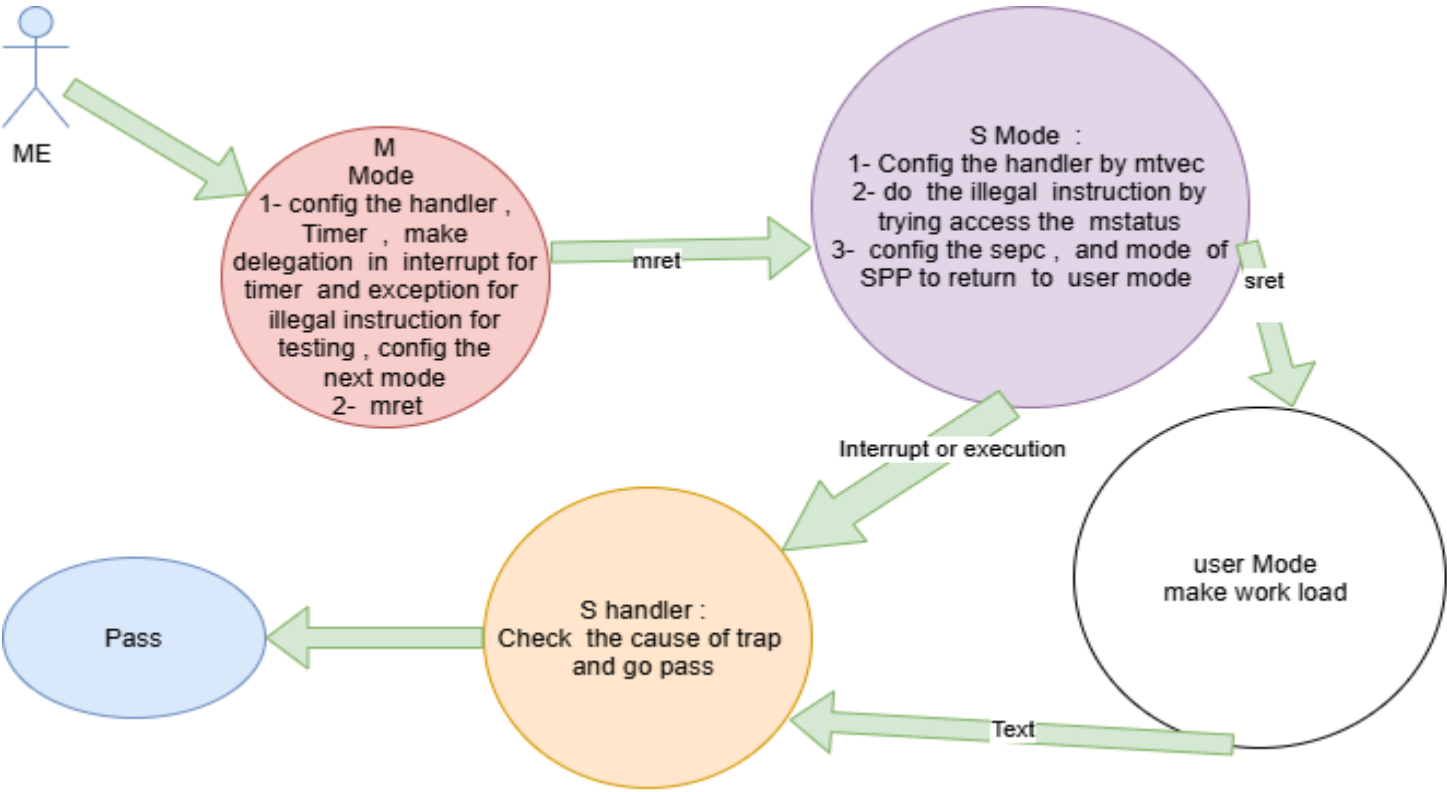


Output of spike :

```
0: >>>> s_mode
0: 0x0000000080000088 (0x00000297) auipc t0, 0x0
0: 1 0x0000000080000088 (0x00000297) x5 0x0000000080000088
0: 0x000000008000008c (0x01428293) addi t0, t0, 20
0: 1 0x000000008000008c (0x01428293) x5 0x000000008000009c
0: 0x0000000080000090 (0x10529073) csrwr stvec, t0
0: 1 0x0000000080000090 (0x10529073) c261_stvec 0x000000008000009c
0: 0x0000000080000094 (0x30002373) csrr t1, mstatus
0: exception trap_illegal_instruction, epc 0x0000000080000094
0: tval 0x0000000030002373
0: >>>> s_priv
0: 0x000000008000009c (0x142022f3) csrr t0, scause
0: 1 0x000000008000009c (0x142022f3) x5 0x0000000000000002
0: 0x00000000800000a0 (0x00004309) c.li t1, 2
0: 1 0x00000000800000a0 (0x4309) x6 0x0000000000000002
0: 0x00000000800000a2 (0x02629563) bne t0, t1, pc + 42
0: 1 0x00000000800000a2 (0x02629563)
0: 0x00000000800000a6 (0x14302bf3) csrr s7, stval
0: 1 0x00000000800000a6 (0x14302bf3) x23 0x0000000030002373
0: 0x00000000800000aa (0x14102c73) csrr s8, sepc
0: 1 0x00000000800000aa (0x14102c73) x24 0x0000000080000094
0: 0x00000000800000ae (0x00004515) c.li a0, 5
0: 1 0x00000000800000ae (0x4515) x10 0x0000000000000005
0: 0x00000000800000b0 (0x0000a821) c.j pc + 24
0: 1 0x00000000800000b0 (0xa821)
```

Task2:

Diagram Flow:

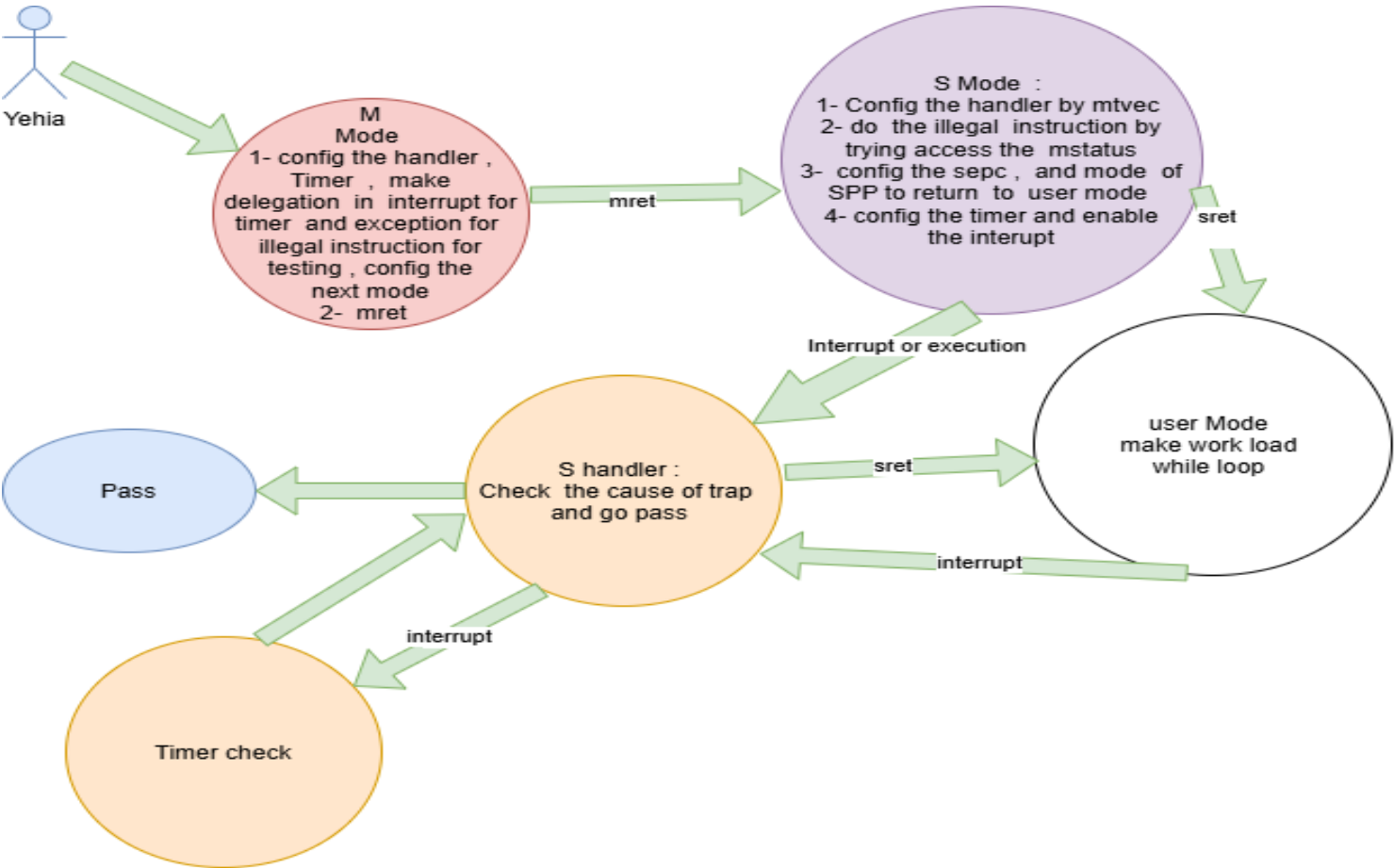


Output of spike:

```
0: 0x00000000800000a8 (0x10020073) csrr sstatus, t0
0: 1 0x00000000800000a8 (0x1002b073) c768_mstatus 0x0000000a00000080
0: 0x00000000800000ac (0x02000293) li t0, 32
0: 1 0x00000000800000ac (0x02000293) x5 0x0000000000000020
0: 0x00000000800000b0 (0x1002a073) csrs sstatus, t0
0: 1 0x00000000800000b0 (0x1002a073) c768_mstatus 0x0000000a000000a0
0: 0x00000000800000b4 (0x10002973) csrr s2, sstatus
0: 1 0x00000000800000b4 (0x10002973) x18 0x0000000200000020
0: 0x00000000800000b8 (0x10200073) sret
0: 1 0x00000000800000b8 (0x10200073) c768_mstatus 0x0000000a000000a2
0: >>>> u_mode_user_app
0: 0x00000000800000c0 (0x00004281) c.li t0, 0
0: 0 0x00000000800000c0 (0x4281) x5 0x0000000000000000
0: 0x00000000800000c2 (0x100022f3) csrr t0, sstatus
0: exception trap_illegal_instruction, epc 0x00000000800000c2
0: tval 0x00000000100022f3
0: >>>> s_mode_trap_handler
0: 0x00000000800000dc (0x142022f3) csrr t0, scause
0: 1 0x00000000800000dc (0x142022f3) x5 0x0000000000000002
0: 0x00000000800000e0 (0x00628763) beq t0, t1, pc + 14
0: 1 0x00000000800000e0 (0x00628763)
0: 0x00000000800000e4 (0x00004321) c.li t1, 8
0: 1 0x00000000800000e4 (0x4321) x6 0x0000000000000008
0: 0x00000000800000e6 (0x02628363) beq t0, t1, pc + 38
0: 1 0x00000000800000e6 (0x02628363)
0: 0x00000000800000ea (0x00008d16) c.mv s10, t0
0: 1 0x00000000800000ea (0x8d16) x26 0x0000000000000002
0: 0x00000000800000ec (0x0000a8b1) c.j pc + 92
0: 1 0x00000000800000ec (0xa8b1)
```

Task 3:

Diagram Flow:



Output of spike:

```
core 0: 0x00000000800000d0 (0x0000bffd) c.j pc - 2
core 0: 0 0x00000000800000d0 (0xbffd)
core 0: 0x00000000800000ce (0x00000505) c.addi a0, 1
core 0: 0 0x00000000800000ce (0x0505) x10 0x00000000000009a4
core 0: 0x00000000800000d0 (0x0000bffd) c.j pc - 2
core 0: 0 0x00000000800000d0 (0xbffd)
core 0: 0x00000000800000ce (0x00000505) c.addi a0, 1
core 0: 0 0x00000000800000ce (0x0505) x10 0x00000000000009a5
core 0: 0x00000000800000d0 (0x0000bffd) c.j pc - 2
core 0: 0 0x00000000800000d0 (0xbffd)
core 0: exception interrupt #5, epc 0x00000000800000ce
core 0: >>>> s_mode_trap_handler
core 0: 0x00000000800000d4 (0x142022f3) csrr t0, scause
core 0: 1 0x00000000800000d4 (0x142022f3) x5 0x8000000000000005
core 0: 0x00000000800000d8 (0x0202c263) bltz t0, pc + 36
core 0: 1 0x00000000800000d8 (0x0202c263)
core 0: >>>> stimer_check
core 0: 0x00000000800000fc (0x00000405) c.addi s0, 1
core 0: 1 0x00000000800000fc (0x0405) x8 0x0000000000000001
core 0: 0x00000000800000fe (0x00000001) c.nop
core 0: 1 0x00000000800000fe (0x0001)
```

The final Questions :

- 1- Without the delegation to trap to s- mode we need to go to m mode and consume instruction in scheduling then transfer to S-mode , there is trade off without the delegation simple hardware but consume time in scheduling each tick , with delegation config the next mode that i want to transfer but we need the sstc extension.
- 2- These are different bits in cause register , MTI --> 7 , STI ---> 5
- 3- Illegal instruction exception with coded 2 , in task 2 and task 1 , i appended this case in testing to verify the transfer concept